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# Computer Science & Information Technology

## Compiler Design

DPP: 1

## Parsing

**Q1** Consider the following grammar G:

**G:** S → [S | SS| ]

Where S is start variable, Number of terminals in follow set of S is_______

**Q2** Consider the following grammar.

S → (S) | SS | ∈

Total number of terminals in follow set of S is___.

**Q3** Consider the following Grammar G:

S → ACB

A → a | b | ∈

B → c | d | ∈

C → e | f

How many terminals are present in First (S)?

(A) 6 (B) 3

(C) 7 (D) 4

**Q4** Which of the following is/are correct regarding FIRST & FOLLOW of the given grammar.

E → TE'

E' → +TE' |∈

T → FT'

T' → *FT' |∈

F → id |(E)

(A) FIRST (T)= {id, (}
FOLLOW(T') = {+, $, )}

(B) FIRST (E) = {id, (}
FOLLOW(E') = {$, )}

(C) FIRST (E) = {id, (}
FOLLOW(F) = {+, $ ,), *}

(D) FIRST (T') = {*, Î}
FOLLOW (E) = {$, * , )}

**Q5** Consider the following grammar G:

G:

S → AaAb | BbBa

A → ∈

B → ∈

Number of terminals in First (S) is/are ____.

**Q6** Consider the following grammar:

S → P R Q | R b Q | Q a

P → d a | Q R

Q → g | ∈

R → h | ∈

The number of elements in Follow (R) and the follows set of R respectively is?

(A) 4, {$, b, g, h}

(B) 3, {b, g, h}

(C) 5, {$, a, b, g, h}

(D) 3, {$, b, g}

**Q7** Which of the following are present in FIRST(S) ∩ FIRST(F)?

S → E

E → Ff | Gh

F → ef | gh | ∈

G → g | ∈

(A) {e, f, g, h}

(B) {e, f, h, ∈}

(C) {e, g, ∈}

(D) {e, g}

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# Answer Key

| | |
|---|---|
| Q1 | 3~3 |
| Q2 | 3~3 |
| Q3 | (D) |
| Q4 | (A, B) |
| Q5 | 3~3 |
| Q6 | (A) |
| Q7 | (D) |

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# Hints & Solutions

**Q1 Text Solution:**

Follow (s) = first (S)

**=** { [, ], $ }

Total 3 terminals in follow (S).

**Q2 Text Solution:**

S → (S) | SS | ε

Follow (S) = {$, ), (}.

Number of non – terminals = 3

Hence, 3 is correct answer.

**Q3 Text Solution:**

First (S) = First (ACB)

First (S) = {a, b} U First (C)

= {a, b, e,f}

Number of terminals = 4

**Q4 Text Solution:**

Given that

| | **Frist** | **Follow** |
|---|---|---|
| E – | {(, id} | {$, )} |
| E′ – | {+, Î} | {$ , )} |
| T – | {id, (} | {+, $ , ) } |
| T′ – | {*, Î} | {+ , $ , )} |
| F – | {id, (} | {+, $, )} |

So, option a, b are correct.

**Q5 Text Solution:**

First (S) = {a,b}

Number of terminals = 2

**Q6 Text Solution:**

First (R) ⇒ {h, $\varepsilon$}

Follow (R) ⇒ First(Q) ∪ {b} ∪ Follow(P)

First (Q) ⇒ {g, $\varepsilon$}

Follow (P) ⇒ First (R) ∪ First (Q) ∪ Follow (S)

⇒ {h} ∪ {g} ∪ {$}

Follow (R) = {g} ∪ {b} ∪ {h, g, $}

= {b, h, g, $}.

**Q7 Text Solution:**

First of all non-terminals.

| | |
|---|---|
| S | {e, f, h, g} |
| E | {e, f, g, h} |
| F | {e, g, ∈} |
| G | {g, ∈} |

So, FIRST(S) ∩ FIRST (F) = {e, f, h, g} ∩ {e, g,∈ } = {e, g}

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